

Overfitting and Underfitting Across Dataset Complexity: A Comparative Study on Auto MPG and NYC Taxi

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Abstract—The phenomena of overfitting and underfitting are fundamental challenges in supervised machine learning, but their manifestation depends strongly on the complexity of the underlying dataset. In this paper we present a comparative empirical study of these failure modes on two regression benchmarks of contrasting complexity: the Auto MPG dataset (392 samples, 7 features, strong linear signal) and the NYC Green Taxi trip-duration dataset (15 000 samples, 7 features, weak linear signal and high irreducible noise). We train four feedforward neural networks of varying capacity on each dataset and evaluate them using a single train/test split, 10-fold cross-validation, learning curves, and a hyperparameter sensitivity analysis. Results show that overfitting emerges at much lower model capacities on the simple Auto MPG dataset, while the complex taxi dataset exhibits weaker overfitting signals and substantially higher irreducible noise. L_2 regularisation is highly effective on Auto MPG but yields comparatively modest improvements on the taxi data. These findings highlight that the capacity threshold for generalisation failure is not a fixed property of the model family, but depends critically on dataset complexity, sample size, and the signal-to-noise ratio of the task.

Index Terms—overfitting, underfitting, bias–variance trade-off, neural networks, regression, regularisation, Auto MPG, NYC Taxi

I. INTRODUCTION

Supervised learning models must balance two competing objectives: fitting the training data accurately and generalising to unseen examples. A model with insufficient capacity underfits both training and test data, while a model with excessive capacity memorises noise and overfits [1]. This tension, formalised as the bias–variance trade-off [2], is well understood in theory. However, the practical consequences of model complexity are strongly modulated by properties of the dataset itself, such as sample size, dimensionality, feature collinearity, and the fraction of variance that is irreducible noise.

Despite the theoretical clarity of the bias–variance decomposition, empirical comparisons of overfitting and underfitting across datasets of different complexity are rare in introductory literature. Most demonstrations use a single synthetic or low-complexity dataset, which may give a misleading impression of how sharply these phenomena manifest in practice.

This paper addresses this gap by training identical neural network architectures on two real-world regression benchmarks of contrasting complexity. The Auto MPG dataset [9]

is a canonical low-complexity benchmark with strong linear structure, few samples, and low noise. The NYC Green Taxi trip-duration dataset [11] presents a more complex task: trip duration depends on spatial structure, temporal patterns, and traffic conditions that are only partially captured by the available features, resulting in substantially higher irreducible noise.

The contributions of this paper are:

- an exploratory data analysis of both datasets highlighting their structural differences;
- controlled overfitting and underfitting experiments using four architectures on each dataset;
- a direct comparison of generalisation behaviour across dataset complexity regimes; and
- a fully reproducible Python implementation.

II. BACKGROUND

A. The Bias–Variance Trade-Off

The expected generalisation error of a model $\hat{f}$ decomposes as [2]:

$$\mathbb{E}[(y - \hat{f}(x))^2] = \text{Bias}^2[\hat{f}] + \text{Var}[\hat{f}] + \sigma^2, \quad (1)$$

where σ^2 is the irreducible noise. A critical but often overlooked implication of Equation 1 is that only σ^2 is dataset-dependent in the classical decomposition; however, in practice the capacity threshold at which a model transitions from underfitting to overfitting also depends on the signal-to-noise ratio and sample size of the data [3]. On datasets with high σ^2 , the model must be large enough to capture genuine structure before variance from overfitting becomes the dominant error term.

B. An Information-Theoretic View of Overfitting

Bashir et al. [16] provide a formal information-theoretic framework for understanding overfitting and underfitting. They define the *algorithm capacity* C_A as the maximum mutual information an algorithm can extract from a dataset when selecting its output hypothesis:

$$C_A = \sup_{\mathcal{D}} I(G; D), \quad (2)$$

where G is the random variable representing the algorithm's output hypothesis and D is the training dataset. Under this framework, overfitting is defined as the condition $C_{A,D} > \mathbb{E}[C_D]$: the algorithm extracts more bits from the dataset than are required to represent the true underlying signal. Underfitting is the converse, $C_A^i < \mathbb{E}[C_D]$, where the model has insufficient capacity after i training steps.

Critically, Bashir et al. also prove that the general problem of determining whether an arbitrary algorithm will overfit a given dataset is formally *undecidable*, by reduction from the halting problem [16]. This result implies that no universal, analytical pre-training check can replace empirical evaluation, motivating the diagnostic approach taken in this paper.

C. Feedforward Neural Networks

A feedforward neural network with L hidden layers computes $\hat{f}(x) = W_L \phi(\dots \phi(W_1 x + b_1) \dots) + b_L$, where W_ℓ, b_ℓ are layer weights and biases, and ϕ is a pointwise activation. The universal approximation theorem guarantees that sufficient width can represent any continuous function [4], but capacity must be matched to the task.

As the number of parameters grows relative to the number of training samples, the network can interpolate the training data exactly, which is rarely beneficial for generalisation [5].

D. Regularisation

L_2 weight decay adds $\lambda \|W\|_2^2$ to the loss, shrinking weights and discouraging noise memorisation [1]. Its effectiveness depends on the structure of the problem: on low-noise datasets with smooth target functions, it strongly improves generalisation, while on high-noise tasks it provides more modest gains because a large fraction of the test error is irreducible [3].

E. Related Work

Geman et al. [2] formalised the bias–variance trade-off for neural networks; Hastie et al. [3] provide a comprehensive statistical learning treatment. Zhang et al. [5] showed that modern deep networks can memorise randomly labelled data, demonstrating that classical regularisation theory does not fully explain generalisation. More recently, the double-descent phenomenon [6] has shown that heavily overparameterised models can actually *improve* on test data beyond the classical bias–variance optimum, an effect more pronounced on larger, noisier datasets.

III. DATASETS AND EXPLORATORY ANALYSIS

A. Auto MPG

The Auto MPG dataset [9], [10] contains 398 automobile records; after removing six entries with missing horsepower values, 392 clean samples remain. Seven attributes describe each car: cylinders, displacement, horsepower, weight, acceleration, model year, and origin. The target is fuel efficiency in miles per gallon (MPG).

Figure 1 shows the MPG distribution and correlation matrix. The distribution is right-skewed and spans roughly

9–47 MPG. The correlation matrix reveals strong multi-collinearity: weight, displacement, cylinders, and horsepower share pairwise correlations above 0.8, and all four are strongly negatively correlated with MPG. This means that even a model exploiting only one of these features can achieve reasonable performance, which explains why the signal is strong and noise is low.

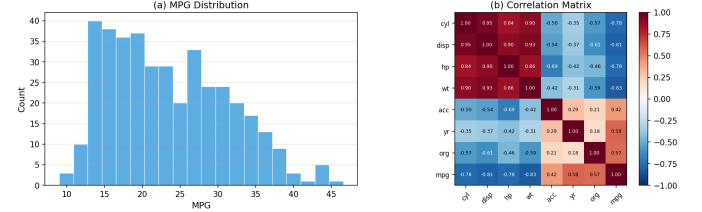


Fig. 1. Auto MPG: (a) target distribution; (b) Pearson correlation matrix. Strong multi-collinearity is visible among physical attributes.

Figure 2 shows the absolute Pearson correlation with MPG. Weight, displacement, cylinders, and horsepower dominate; origin and acceleration contribute least individually.

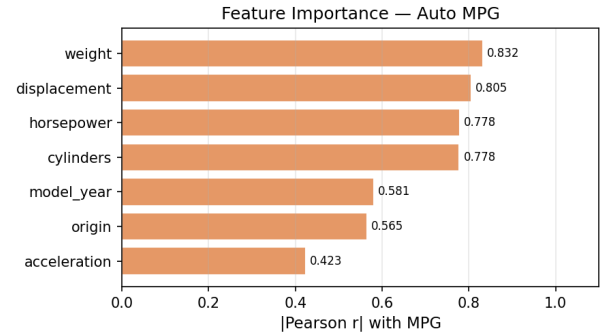


Fig. 2. Auto MPG: absolute Pearson correlation of each feature with the target.

B. NYC Green Taxi Trip Duration

The NYC Taxi and Limousine Commission (TLC) publishes monthly trip records for all licensed taxis [11]. We use the January 2023 green taxi dataset, from which we compute trip duration as the difference between drop-off and pick-up timestamps. After filtering for durations between 1 and 120 minutes and non-zero trip distances, we sample 15 000 records at random for computational tractability.

Features used are: trip distance, passenger count, pick-up and drop-off location IDs (integers identifying NYC taxi zones), pick-up hour, day of week, and rate code. Unlike Auto MPG, location IDs encode spatial information only implicitly as integer codes, and many factors affecting trip duration—traffic incidents, signal timing, weather—are absent from the data entirely, resulting in high irreducible noise.

Figure 3 shows the duration distribution and correlation matrix. The distribution is strongly right-skewed with a long tail, and correlations with the target are substantially weaker than in Auto MPG. Trip distance is the only feature with a

moderate linear relationship to duration; the remaining features are nearly uncorrelated with the target individually. This low linear signal is a key structural difference from Auto MPG.

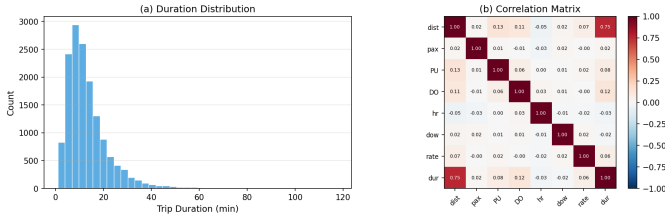


Fig. 3. NYC Taxi: (a) trip duration distribution; (b) Pearson correlation matrix. Correlations with the target are considerably weaker than in Auto MPG.

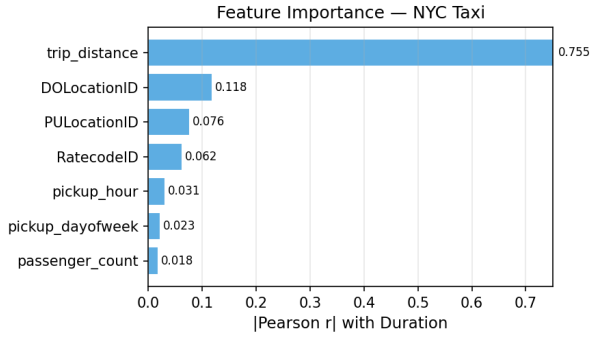


Fig. 4. NYC Taxi: absolute Pearson correlation of each feature with trip duration. Only trip distance shows a moderate linear relationship.

IV. METHODOLOGY

A. Data Preprocessing

Each dataset is split 80/20 into training and test sets with a fixed random seed. Input features are standardised to zero mean and unit variance using statistics estimated from the training split only.

B. Models

Four MLP architectures are trained on each dataset (Table I). All models use the ReLU activation, Adam optimiser [7], and up to 5 000 training epochs. The regularised model uses the same architecture as the overfitting model but with L_2 coefficient $\lambda = 0.1$.

TABLE I
MODEL ARCHITECTURES. ALL MODELS SHARE THE SAME HYPERPARAMETERS EXCEPT WHERE NOTED.

Model	Layers	Units	Params	λ
Underfitting	1	2	≈ 17	10^{-4}
Good Fit	2	16	≈ 545	10^{-4}
Overfitting	5	64	$\approx 20k$	10^{-4}
Regularised	5	64	$\approx 20k$	0.1

C. Evaluation

Each model is evaluated by four protocols: (i) single train/test split MSE and predicted-vs-actual scatter plots; (ii) training loss curves for convergence analysis; (iii) 10-fold cross-validation MSE with standard deviation; (iv) learning curves (training set size 10%–100%, 5-fold CV); and (v) hyperparameter sensitivity (units per layer $\in \{2, 4, 8, 16, 32, 64, 128\}$, 2-layer network). To compare datasets directly, the train–test MSE gap is normalised by the target variance of each dataset. All code uses `scikit-learn` [12], `pandas` [13], `NumPy` [14], and `Matplotlib` [15].

V. RESULTS: AUTO MPG

A. Predictions and MSE

Figure 5 shows predicted vs. actual MPG. The underfitting model collapses predictions to a near-constant value. The good-fit model clusters tightly around the diagonal. The overfitting model fits training points well but scatters on test points, particularly at high MPG values. The regularised model recovers near good-fit behaviour despite its large capacity. Table II quantifies these observations: the overfitting model has the lowest training MSE but the highest test MSE, and regularisation substantially closes the gap.

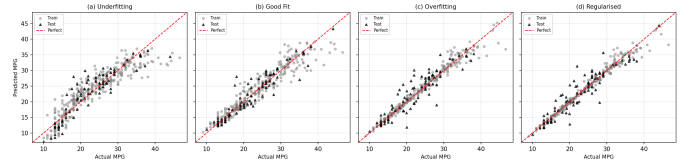


Fig. 5. Auto MPG: predicted vs. actual for the four models. Red dashes = perfect prediction. Grey circles = train, black triangles = test.

TABLE II
AUTO MPG: SINGLE-SPLIT MSE (MPG²).

Model	Train MSE	Test MSE
Underfitting	10.998	10.977
Good Fit	5.733	6.438
Overfitting	1.794	9.345
Regularised	1.297	8.190

MSE values obtained by the experiment.

B. Loss Curves and Cross-Validation

Figure 6 shows training loss curves. The underfitting model plateaus early at a high loss. The overfitting model achieves the lowest training loss with the noisiest trajectory. 10-fold CV results (Figure 7) confirm the single-split findings with tighter bounds, and show that the overfitting model exhibits the largest variance across folds.

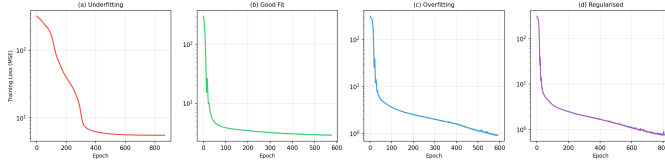


Fig. 6. Auto MPG: training loss curves (log scale).

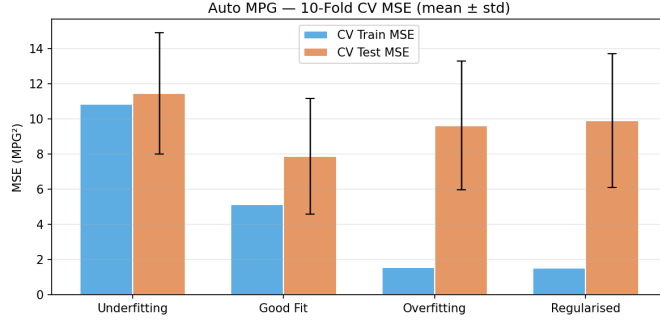


Fig. 7. Auto MPG: 10-fold CV MSE (mean $\pm$ std).

C. Learning Curves and Hyperparameter Sensitivity

Figure 8 shows that the underfitting model is unaffected by additional data (flat curves), while the overfitting model’s train–test gap persists even with the full dataset. The hyperparameter sensitivity curve (Figure 9) shows a clear U-shaped test MSE with a minimum around 16 units per layer, confirming a well-defined optimal capacity for this dataset.

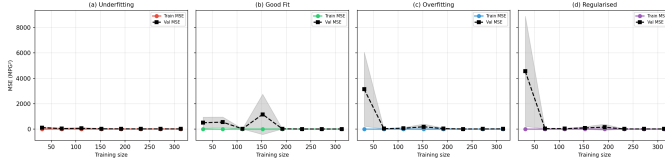


Fig. 8. Auto MPG: learning curves. The underfitting model (a) is insensitive to data volume; the overfitting model (c) retains a persistent gap.

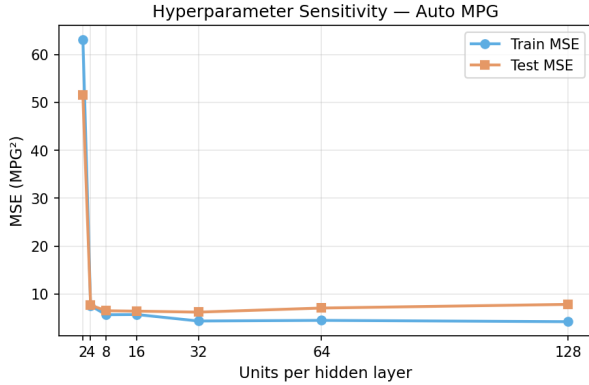


Fig. 9. Auto MPG: hyperparameter sensitivity. Test MSE follows a clear U-shaped curve with a minimum near 16 units per layer.

VI. RESULTS: NYC TAXI

A. Predictions and MSE

Figure 10 shows predicted vs. actual trip duration. Even the good-fit model shows substantially more scatter than its Auto MPG counterpart, reflecting the higher irreducible noise. The overfitting model fits training points more closely but the test scatter is only marginally worse, in contrast to the dramatic gap seen on Auto MPG. This is consistent with the weaker signal in the taxi data: there is less structure to memorise. Table III shows that the absolute MSE values are considerably higher than Auto MPG, and the train–test gap for the overfitting model is narrower in relative terms.

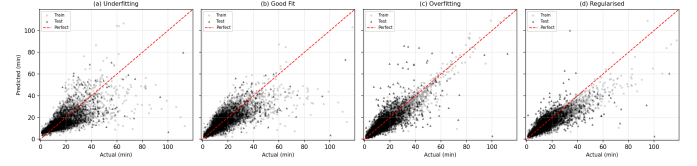


Fig. 10. NYC Taxi: predicted vs. actual trip duration for the four models.

TABLE III
NYC TAXI: SINGLE-SPLIT MSE (MIN²).

Model	Train MSE	Test MSE
Underfitting	37.303	37.151
Good Fit	28.509	30.917
Overfitting	12.048	39.130
Regularised	17.871	36.063

MSE Values obtained by the experiment.

B. Loss Curves and Cross-Validation

Figure 11 shows that all models require more epochs to converge on taxi data. The good-fit and overfitting models converge to similar training loss, suggesting that the additional capacity of the overfitting model provides diminishing returns on a noisy task. CV results (Figure 12) show wider error bars than Auto MPG for all models, reflecting the higher difficulty of the task.

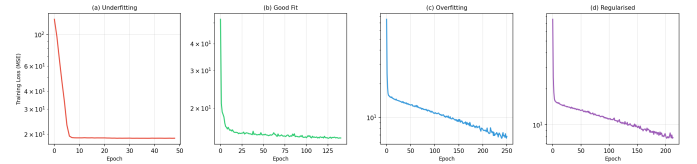


Fig. 11. NYC Taxi: training loss curves (log scale).

C. Learning Curves and Hyperparameter Sensitivity

Figure 13 shows that all non-trivial models benefit from additional training data on the taxi task, with the train–test gap closing steadily — the opposite of the flat underfitting behaviour and persistent overfitting gap seen on Auto MPG.

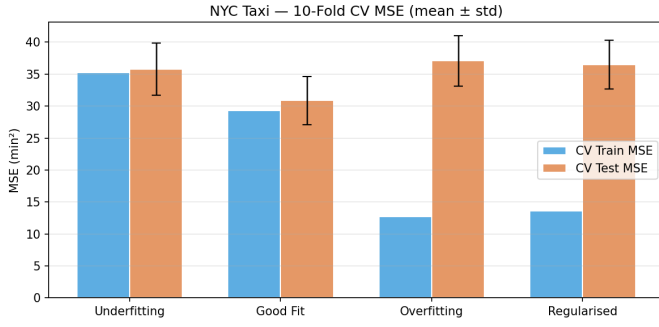


Fig. 12. NYC Taxi: 10-fold CV MSE (mean ± std).

This suggests that the taxi dataset is data-hungry: more samples would likely continue to improve generalisation. The hyperparameter sensitivity curve (Figure 14) is noticeably flatter than for Auto MPG, without a sharp U-shape. Test MSE decreases gradually as capacity increases, indicating that the dataset complexity prevents any architecture from memorising the training data as effectively.

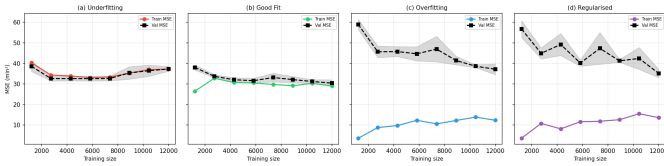


Fig. 13. NYC Taxi: learning curves. All models continue to improve with more data, unlike Auto MPG.

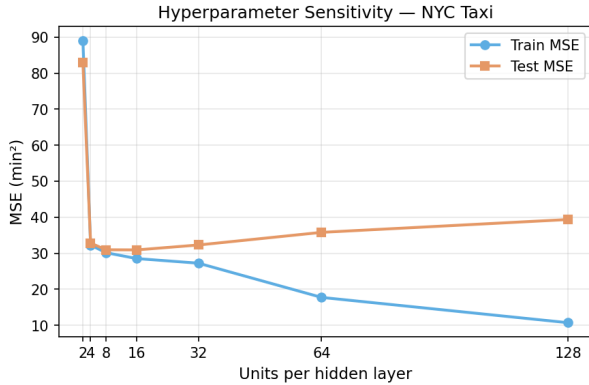


Fig. 14. NYC Taxi: hyperparameter sensitivity. The test MSE curve is flatter than on Auto MPG and lacks a clear minimum.

VII. COMPARATIVE DISCUSSION

A. Generalisation Gap Across Datasets

Figure 15 shows the normalised train–test MSE gap ($(MSE_{test} - MSE_{train}) / \text{Var}(y)$) for both datasets. On Auto MPG, the overfitting model’s normalised gap is substantially larger than on NYC Taxi, confirming that sharp overfitting is characteristic of low-complexity, low-noise data where the

model can successfully memorise training patterns that do not generalise.

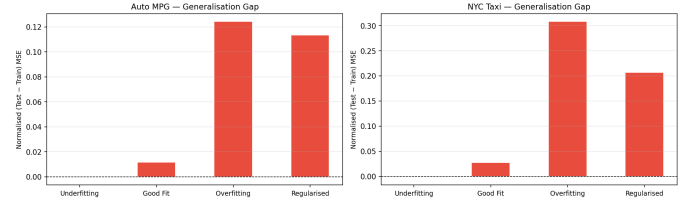


Fig. 15. Normalised (test – train) MSE gap for each model on both datasets. Positive bars (red) indicate overfitting; negative or near-zero bars (green) indicate underfitting or good fit.

B. Hyperparameter Sensitivity Comparison

Figure 16 juxtaposes the hyperparameter sensitivity curves. Auto MPG shows a pronounced U-shape with a clear optimal capacity, while NYC Taxi shows a much flatter test MSE curve. This difference has practical implications: model selection on simple datasets is more sensitive to capacity choice, and a poorly chosen architecture can dramatically degrade performance. On complex, noisy datasets the penalty for excess capacity is smaller, but the reward for finding the right capacity is also lower because the floor is set by irreducible noise.

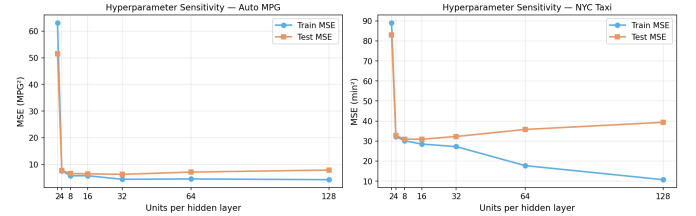


Fig. 16. Hyperparameter sensitivity comparison. The U-shaped minimum is sharp on Auto MPG but flat on NYC Taxi.

C. Effect of Regularisation

L_2 regularisation has a strong positive effect on Auto MPG, substantially closing the train–test gap. On NYC Taxi the improvement is more modest, consistent with the theoretical prediction that regularisation reduces variance but cannot reduce irreducible noise. When the dominant component of test error is σ^2 rather than variance, regularisation provides limited benefit.

D. Underfitting: Obvious vs. Subtle

On Auto MPG, underfitting is immediately visible in the prediction scatter plots as a near-constant output. On NYC Taxi, underfitting is more subtle: the model produces predictions that vary, but systematically miss the full dynamic range of trip durations and fail on the long-duration tail. This illustrates a practical pitfall: on noisy, complex tasks, underfitting can be mistaken for irreducible noise if only aggregate metrics like MSE are examined.

E. Limitations

The taxi dataset is sampled to 15 000 records for tractability; the full dataset contains millions of trips. At full scale, larger models would likely overfit less due to the abundance of data. Additionally, location IDs are used as raw integers rather than as embeddings, which may understate the true signal in the spatial features. Future work could apply graph-based or embedding-based representations of taxi zones and compare overfitting behaviour with the current setup.

F. Capacity-Complexity Mismatch Across Datasets

The information-theoretic framework of Bashir et al. [16] provides an additional lens through which to interpret the contrasting overfitting behaviour observed on the two datasets. Auto MPG has low dataset complexity C_D : its target function is largely explained by a small set of highly correlated features, and the true signal is compressible. Even a moderately-sized network can extract more bits from the training set than the signal warrants, producing capacity-complexity mismatch and sharp overfitting. The NYC Taxi dataset has substantially higher C_D : traffic dynamics, signal timing, and spatial structure contribute to a complex and partially unobservable target function. The capacity threshold at which $C_{A,D}$ begins to exceed $\mathbb{E}[C_D]$ is therefore higher, consistent with the weaker and flatter overfitting signal observed empirically.

VIII. CONCLUSION

We have presented a comparative empirical study of overfitting and underfitting on two regression datasets of contrasting complexity. Our experiments show that the capacity threshold at which a neural network begins to overfit is not a fixed property of the model architecture, but depends on the dataset's signal-to-noise ratio, sample size, and feature structure. On the simple Auto MPG dataset, overfitting is sharp, easily diagnosed, and effectively corrected by L_2 regularisation. On the more complex NYC Taxi dataset, the overfitting signal is weaker, the hyperparameter optimum is flatter, and the dominant challenge is irreducible noise rather than model variance. These findings underscore the importance of characterising dataset complexity before making model selection decisions.

Several directions remain open for future work. First, the NYC Taxi experiments use a sample of 15 000 records drawn from a dataset of several million trips. Replicating the analysis at larger scales would clarify whether the weak overfitting signal is a consequence of dataset complexity alone or is also partly due to the limited sample used here; at full scale, the volume of data would itself act as a regulariser and the train-test gap for the overfitting model would likely narrow further. Second, only L_2 weight decay was studied as a regularisation strategy. Comparing dropout [8] and early stopping in the same controlled setting would give a more complete picture of which techniques are most effective in each complexity regime. Third, taxi zone location IDs are currently encoded as raw integers, which does not reflect the spatial relationships between zones. Replacing them with learned embeddings or

graph-based representations could substantially reduce irreducible noise on the taxi task and make the overfitting-underfitting contrast between the two datasets sharper. Finally, extending the framework to a third dataset of intermediate complexity—such as the California Housing dataset—would allow the capacity-complexity relationship identified in this paper to be studied as a continuum rather than a binary comparison, and would provide an opportunity to empirically estimate algorithm capacity C_A following the information-theoretic framework of Bashir et al. [16].

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